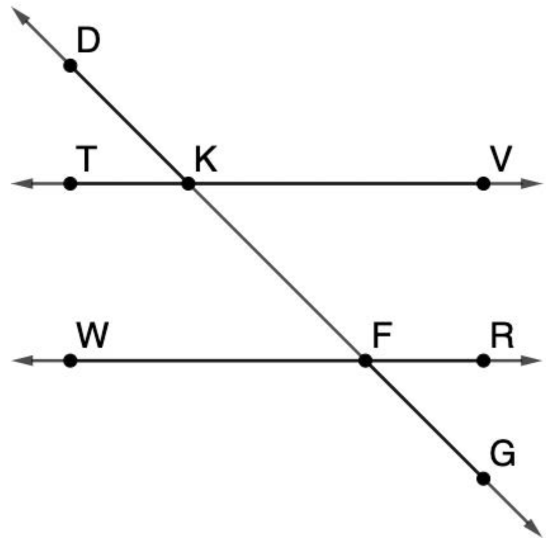
Geometry
Unit 3
Lesson 9 Homework

Name Answer Key_________
Date____________________
Period___________________

Complete the two-column proof.

Given: $\angle TKF \cong \angle KFR$

Prove: $\angle VKD \cong \angle WFG$



Statement	Reason
$\angle TKF \cong \angle KFR$	Given
1. $\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	2. If alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel.
3. $\angle VKD \cong \angle WFG$	4. If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.

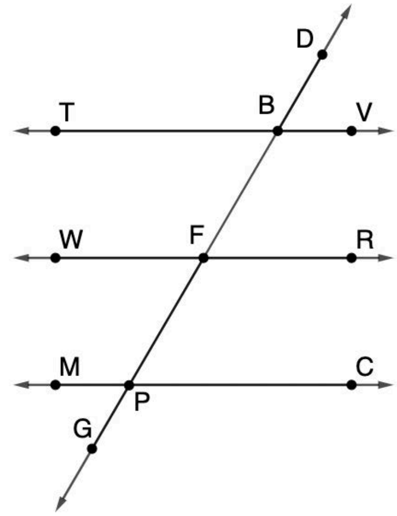
Write the statements and reasons in the correct order in the two-column proof.

$\angle TKD \cong \angle GFR$	Supplements of congruent angles are congruent.
$\angle VKD \cong \angle WFG$	If two angles form a linear pair, then they are supplementary
$\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle GFR$ and $\angle WFG$ are supplementary	If alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.

Given: $\angle DBV \cong \angle MPG$ and $\angle TBD \cong \angle WFB$

Prove: $\angle RFP \cong \angle MPF$



Statement	**Reason**
$\angle DBV \cong \angle MPG$	Given
5. $\overleftrightarrow{TV} \parallel \overleftrightarrow{MC}$	5. If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
6. $\angle TBD \cong \angle WFB$	6. Given
7. $\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	7. If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.
8. $\angle PFR \cong \angle TBD$	8. If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
9. $\angle TBD \cong \angle MPF$	9. If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
10. $\angle RFP \cong \angle MPF$	10. Transitive Property of Congruence

Write the statements and reasons in the correct order in the two-column proof.

$\angle RFP \cong \angle MPF$	Given
$\angle TBD \cong \angle WFB$	Transitive Property of Congruence
$\angle PFR \cong \angle TBD$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle TBD \cong \angle MPF$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\overleftrightarrow{TV} \parallel \overleftrightarrow{MC}$	If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
$\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.